Candidate-pool opportunity and selection stability under finite validation in molecular property prediction: a repeated nested audit

Additional file 1

Supplementary Methods and Results

S1 Evidence hierarchy and reconstruction

Evidence was retained in five levels that answer different questions and are not combined into one leaderboard. The current primary analysis is the nine-endpoint, 32-candidate, ten-seed lightweight-registry audit reported in Section S22 and Tables S37–S46. The original effective-diversity, mechanism, multiview, registry-composition, reliability and chemical-support analyses in Sections S1–S16 retain five seeds where ten-seed outputs do not exist and are explicitly secondary. Historical TDC, AutoGluon, conformal, MoleculeACE, bRo5, deduplication and failure-case records remain contextual or negative evidence rather than independent confirmation.

S2 Seed, split and prediction-hash audit

The current primary lightweight registry used ten distinct seed-specific scaffold partitions: 11, 23, 37, 53, 71, 83, 97, 113, 127 and 149. The original supplementary panels in Sections S3–S16 retain seeds 11, 23, 37, 53 and 71 unless a ten-seed source is explicitly cited. Regression Bemis–Murcko groups in the current primary matrix were allocated intact with seed-dependent tie breaking and sample-count balancing; legacy unseeded GroupKFold files are restricted to the split-transition provenance audit.

S3 Effective-rank bootstrap and reference sensitivity

The ten-seed primary effective-diversity analysis used raw, row-centred, fixed-reference-relative and within-unit-rank matrices. At K = 32, endpoint-median Ledoit–Wolf entropy ranks were 2.36, 19.25, 5.56 and 24.33, respectively. These values reflect matrix construction rather than independent candidate counts. Tables S6–S7 provide complete 30 × K outer and 90 × K inner results, leave-one-seed/fold sensitivities and prespecified-reference analyses; no five-seed effective-rank row is primary.

S4 Chance-adjusted ranking aggregation

The original chance-adjusted ranking source contains 540 endpoint–split-seed–outer-fold–K rows from nine endpoints, five split seeds, three outer folds and four K values and is therefore a five-seed secondary analysis. The ten-seed primary reference-gap and practical-equivalence units contain 1,080 rows and are reported separately in Tables S37–S38. No five-seed ranking estimate is presented as a ten-seed primary result.

S5.1 Mechanism-calibration controls

The random-rank negative control used 5,000 replicates and repeated the empirical fold-to-seed-to-endpoint aggregation. The graded positive control used 4,320 locked outer-candidate utilities from 135 audit units, four K values and six injected validation–audit signal levels; non-perfect cells used 500 simulations. Observed CAHit@3 exceeded the 95% permutation envelope at every K. CAHit@3 and normalized MRR gain increased monotonically, while fixed-range selection loss decreased monotonically, with injected signal at every K. Tables S21-S22 retain every summary and outer-unit value.

S5.2 Cross-fitted intervals

In the original five-seed secondary cross-fitted analysis, the three outer folds were averaged within each of five split seeds before resampling. The current primary K-invariant contrasts instead average folds within ten seed blocks and use the other nine seeds to select the held-out reference. Classification ROC-AUC and regression RMSE remain separate evidence strata in both analyses; their units are not pooled.

Effective diversity is an exception to the retained five-seed ranking panels: complete utilities were available for all ten current seeds. Tables S6-S7 therefore use 30 x K outer and 90 x K inner matrices for every endpoint and report raw, row-centred, fixed-reference-relative and within-unit-rank estimates, leave-one-seed/fold sensitivities and predefined-reference analyses. No five-seed effective-rank row is presented as the primary diversity estimate.

S6 Exhaustive matched K = 3 analysis

All C(12,3) = 220 subsets were enumerated from stored candidate-level inner and outer utilities without retraining. The endpoint remains the interpretation unit; overlapping subsets are sensitivity contrasts within one registry rather than independent experiments. Mutually exclusive composition classes, medians, IQRs, distribution ranges and within-endpoint outperforming-Morgan proportions are supplied. CAHit@3 is not estimable because both observed and random Hit@3 equal one at K = 3.

S7 Chemical-support and scaffold-boundary audit

Molecules were stratified by maximum training-set Morgan Tanimoto as <0.5, 0.5 to <0.7 and at least 0.7. Traceable selected-candidate performance, error overlap, disagreement, uncertainty–error association and false-negative or high-error enrichment are reported by support stratum. Novel-scaffold quantities are expressed relative to seen-or-related scaffolds as reference = 1. These analyses are exploratory reliability boundaries rather than independent validation.

S8 TDC endpoint analysis

Twenty-two TDC endpoint summaries retain official metric direction, the historical comparator, performance-run source, retained model and direction-normalized change. Evaluation budgets differ from the primary audit, so these rows provide contextual reliability evidence only.

S9 Automated machine-learning budgets

AutoGluon summaries cover 30-, 300- and 1,800-second budgets across nine datasets, with 27 budget summaries and 81 outer-fold records. Fit time, model count, memory, refit status and comparisons with frozen selector baselines are retained in Table S19. Unequal feature preparation, hardware and search exposure preclude a superiority claim.

S10 Selection-risk and leave-one-endpoint-out analyses

Selection-risk correlations, null calibration and nine leave-one-endpoint-out policy records are retained in Table S20. They are negative or exploratory evidence: nine endpoints cannot identify a generally transferable meta-selector, and endpoint meta-features were evaluated retrospectively.

S11 Conformal prediction and CQR

Table S14 contains 189 classification-conformal and regression-CQR summaries at 80%, 90% and 95% nominal coverage where available. Label-conditional and similarity-conditioned methods improved minority coverage in several tasks but enlarged prediction sets. Coverage remains conditional on the evaluated folds and does not guarantee screening utility after a new chemical shift.

S12 MoleculeACE, bRo5 and deduplication sensitivity

MoleculeACE includes 17 task summaries of cliff-pair direction accuracy, gap association and pair count. The bRo5 panel retains four split summaries, including random, scaffold, perimeter and time settings where available. Eighteen deduplication-policy summaries compare global, training-fold and retained-duplicate strategies. None is presented as independent confirmation of the primary candidate-expansion effect.

S13 Representation errors and failure cases

The limited prediction panel contains RDKit-RF, a GCN and frozen ChemBERTa and MoLFormer probes. Prediction and error correlations were incomplete, and training budgets were unequal. Low-support, novel-scaffold, activity-cliff, bRo5 and ClinTox minority failures delimit the evidence rather than establish a modern-model leaderboard.

S14 Split-regime transfer audit

The transfer audit retained the locked candidate registry, K prefixes, five split seeds and 3 x 3 nested design for ClinTox, BACE and ESOL. New structure-separated folds used intact connected components of a Morgan-512 similarity graph with edges at Tanimoto >= 0.70. All 5,760 new fits completed without fallback. CAHit@3 K32 - K4 changes and cross-fitted gap directions agreed between scaffold and similarity-cluster regimes in all three endpoints, but magnitude and interval exclusion changed. Effective-rank estimates across endpoint, K and transformation combinations had Spearman correlation 0.832. Tables S23-S26 and Figure S18 retain the full unit-level results, split manifests and source data.

S15 Expanded registry-composition intervention

The expanded intervention fixed six endpoints (ClinTox, BACE, BBBP, ESOL, Lipophilicity and Caco2_Wang), three locked registries and exact prefixes K = 4, 8, 16 and 32. Five split seeds and three outer folds yielded 1,080 primary outer units. Unit-level records contain 64,616.35 downstream fit/predict seconds; the total expanded-intervention fit count was not reconstructed.

The homogeneous registry used Morgan learner/tuning variants, the classical multiview registry combined Morgan, MACCS, RDKit descriptors and concatenated representations, and the modern-augmented registry interleaved locked frozen ChemBERTa and MoLFormer representations plus a separately locked one-epoch D-MPNN. The D-MPNN candidate belonged only to this expanded intervention and not to the limited four-model prediction panel. Fixed-K component ablations at K = 16 and 32 replaced prespecified classical candidates so that K remained exact.

Anchor sensitivity used three prespecified anchors: the shared Morgan linear candidate, a fixed Morgan random forest and the predefined registry-median candidate. Normalization sensitivity retained raw gains, endpoint-MAD normalization and paired homogeneous-audit-best normalization. The paired normalization was undefined when homogeneous finite-audit opportunity was at most 1e-12; such cells were retained as missing rather than regularized or imputed.

For equal-downstream-budget sensitivity, each prefix was truncated at the endpoint- and K-specific median downstream time of the classical multiview registry; outer performance never determined retention. Recorded downstream time includes inner and outer downstream fit/predict time and excludes model acquisition, encoder pretraining and cached embedding extraction. The Pareto frontier was constructed descriptively by sorting eligible pool–K–design points by mean downstream time and retaining each point only when its mean normalized selected gain exceeded the running maximum. It is a bounded downstream-compute analysis, not an end-to-end efficiency claim.

Selection stability used candidate selection frequencies π_j, normalized selection entropy, modal proportion, leave-one-seed-out agreement and adjacent-fold switches. The three-endpoint split loop repeated all three registries under seeded scaffold and Tanimoto-component splits with endpoint, seed, fold, K, selection and metric definitions held fixed. Complete registries, unit-level records, component results, anchor and normalization sensitivity, split results, stability and budget quantities are provided in Tables S32–S36 and Figures S16–S21.

S16 Source limitations

All-candidate molecule-level predictions are complete for the regression registry and for a post-hoc rerun of the classification 32-candidate registry. The classification rerun showed small stochastic refit drift (maximum absolute outer-candidate metric difference 0.0010 and two one-standard-error selection changes), so locked primary metrics remain the source of record and the new prediction exports are used only for traceability. Complete candidate-level prediction exports remain unavailable across the full multiview registry, so prediction-level diversity is not evaluated for the matched-subset analysis. The public endpoint panels are retrospective, and independently reproduced prospective validation is outside the present study scope.

Supplementary table directory

Table S33. Modern-component ablations and component selection frequencies.

Table S34. Anchor and normalization sensitivity with direction concordance.

Table S35. Composition-by-split loop, direction concordance and split-specific selection stability.

Table S36. Selection stability, candidate frequencies, equal-budget stability and equal-budget diversity.

Supplementary figure directory

Equation-to-code mapping and version provenance. Main-text Equations (5)–(11) define ĵ_u(K), $j_{\mathrm{ref}}^{\mathrm{inv}}(-s)$, $j_{\mathrm{ref}}^{\mathrm{dep}}(-s, K)$, G_{inv}, G_{dep}, G_{same}, G_{avail}, $G_{\mathrm{inv}} = G_{\mathrm{avail}} + G_{\mathrm{dep}}$ and $\Delta_{\mathrm{inv}}(e)$; Equations (12)–(19) continue the matrix, composition and selection-stability definitions. The synchronized mapping is supplied in Additional file 4 under docs/Equation_to_code_mapping.csv. The verified public base remains paper-release-2026-07-r9 at commit

9635a902fa3cc7bb7b71a234c1b2bbbe415193f0; the paper43 ten-seed completion overlay is on public-release hold.

S22. Completion experiments for finite-validation realisation

These analyses complete the current primary lightweight-registry matrix. They were retrospectively specified and subsequently locked, not prospectively registered confirmation, and use the same public endpoint populations and split generator. The full seed set was 11, 23, 37, 53, 71, 83, 97, 113, 127 and 149; the primary outer and inner matrices therefore contain 30 × K and 90 × K rows, respectively.

S22.1 K-invariant reference and practical equivalence

For each held-out seed s, $j_{\mathrm{ref}}^{\mathrm{inv}}(-s)$ was selected once from all 32 candidates using the other nine seeds; $j_{\mathrm{ref}}^{\mathrm{dep}}(-s,K)$ was selected analogously within $\mathcal{C}_K$. For u = (s,f), G_{inv} and G_{dep} subtract the utility of j-hat_u(K) from the respective reference utility, G_{same} subtracts it from the same-fold eligible maximum, and G_{avail} is the K-invariant reference utility minus the K-dependent reference utility. Thus G_{inv} = G_{avail} + G_{dep} exactly. The seed-level mean $\bar{G}_{\mathrm{inv},e,s}(K)$ first averages G_{inv} over the three outer folds. $\Delta_{\mathrm{inv}}(e)$ then averages Gbar_inv,e,s(32) − Gbar_inv,e,s(4) over the ten seed blocks; negative values favour the smaller full-registry completion gap at K = 32. Practical-equivalence tolerance τ grids were 0.005/0.010/0.020 ROC-AUC and 0.025/0.050/0.100 RMSE and remain retrospective reporting tolerances rather than minimum important differences. The distinct numerical-stability constant $\varepsilon_{\mathrm{num}} = 10^{-12}$ is used only for floating-point stabilization and small-denominator missingness.

Table S37. K-invariant K = 32 minus K = 4 effects

Endpoint	Type	Mean contrast	95% low	95% high
bace	classification	-0.0195	-0.0238	-0.0152
bbbp	classification	-0.0263	-0.0301	-0.0225
clintox	classification	-0.0144	-0.0241	-0.0031
esol	regression	0.0683	0.0618	0.0757
freesolv	regression	-0.0586	-0.0912	-0.0263

Endpoint	Type	Mean contrast	95% low	95% high
lipo	regression	-0.0947	-0.0992	-0.0899
tdc_caco2_wang	regression	-0.0049	-0.0121	0.0008
tdc_hia_hou	classification	-0.0022	-0.0075	0.0031
tdc_pgp_broccatelli	classification	-0.0222	-0.0271	-0.0178

S22.2 Metric-matched classification selection

ROC-AUC, PR-AUC and minority-recall-constrained candidate selection were recalculated from inner-fold predictions on the frozen ten-seed splits. Minority status was defined from training labels. Thresholds targeted inner minority recall of at least 0.80 while maximizing majority recall and were aggregated by the inner-fold median. If no candidate met the constraint, the full eligible pool and the locked lexicographic tie rule were used. No additional post-hoc probability calibration was applied; all outcomes were evaluated once on the outer fold.

Table S40. K = 32 selector comparison

Selector	Switch rate	Delta ROC-AUC	Delta PR-AUC	Delta minority recall	Delta majority recall	Delta false-negative rate	Inner feasible rate	Outer target-satisfaction rate
pr_auc	0.5267	-0.0026	0.0018	-0.0002	-0.0136	0.0002	nan	nan
minority_constrained	0.6600	-0.0008	0.0007	-0.0094	0.0059	0.0094	1.0000	0.6200

S22.3 Constructed candidate controls and order sensitivity

Exact duplicates and 95% anchor near-duplicates were constructed from stored score vectors. Weak and complementary-strong candidate orders were learned only from non-held seeds. Best-first, worst-first, cost-first and 100 random permutations assess order sensitivity without additional fitting.

Table S41. Constructed control summary

Control	K	Effective rank	Same-fold gap	Cross-fit gap	Available opportunity
complementary_strong_added	4	1.4046	0.0279	0.0171	0.3193
complementary_strong_added	8	1.7106	0.0449	0.0239	0.3520
complementary_strong_added	16	1.5041	0.0475	0.0235	0.3591
complementary_strong_added	32	1.4455	0.0493	0.0242	0.3601
exact_duplicate	4	1.4046	0.0279	0.0171	0.3193
exact_duplicate	8	1.4046	0.0279	0.0171	0.3193
exact_duplicate	16	1.4046	0.0279	0.0171	0.3193
exact_duplicate	32	1.4046	0.0279	0.0171	0.3193
near_duplicate_95pct_anchor	4	1.4046	0.0279	0.0171	0.3193
near_duplicate_95pct_anchor	8	1.3979	0.0281	0.0172	0.3199
near_duplicate_95pct_anchor	16	1.3945	0.0286	0.0174	0.3204
near_duplicate_95pct_anchor	32	1.3935	0.0288	0.0174	0.3209
registered	4	1.4046	0.0279	0.0171	0.3193

Control	K	Effective rank	Same-fold gap	Cross-fit gap	Available opportunity
registered	8	1.6481	0.0442	0.0256	0.3405
registered	16	1.5713	0.0461	0.0215	0.3525
registered	32	1.4455	0.0493	0.0242	0.3601
weak_added	4	1.4046	0.0279	0.0171	0.3193
weak_added	8	1.7570	0.0362	0.0218	0.3229
weak_added	16	1.6163	0.0480	0.0293	0.3339
weak_added	32	1.4455	0.0493	0.0242	0.3601

S22.4 Correlated heteroskedastic recovery simulation

Validation-minus-audit residual covariance was estimated separately by endpoint. Positive-semidefinite eigenvalue clipping was applied before simulation. Empirical correlated heteroskedastic and independent homoskedastic noise were compared at 0.5, 1 and 2 times the estimated scale, using 2,000 replicates per endpoint, K and scenario.

Table S42. K = 32 empirical recovery summary

Endpoint	Mean simulated gap	95% low	95% high	Top-1 recovery	Observed covered
bace	0.0058	0.0033	0.0089	0.2554	1
bbbp	0.0062	0.0037	0.0091	0.2894	1
clintox	0.0152	0.0102	0.0205	0.2315	0
esol	0.0516	0.0272	0.0848	0.3612	0
freesolv	0.1133	0.0622	0.1762	0.3730	0
lipo	0.0029	0.0011	0.0056	0.6305	0
tdc_caco2_wang	0.0128	0.0074	0.0197	0.2910	0
tdc_hia_hou	0.0154	0.0100	0.0213	0.2202	0
tdc_pgp_broccatelli	0.0067	0.0041	0.0099	0.2887	1

S22.5 Execution and unavailable audits

The unique current ten-seed lightweight registry comprised 34,560 candidate fits and 8,130.16 recorded candidate-fit seconds. The first-five-seed current subset contains 17,280 of these fits; it is not added again. Historical five-seed outputs comprised 17,280 fits and 4,437.95 seconds and remain provenance only. Legacy regression outputs used unseeded GroupKFold and were excluded from the current primary matrix. The maximum legacy-to-current classification difference was 0.000585; the regression split-transition difference reached 1.4217 and is an estimand change, not numerical reproducibility error. Timestamps, independent source-cohort identifiers and exact molecule-level pretraining manifests were unavailable; temporal/source validation was not performed and pretrained-data overlap cannot be excluded.